

[--- CODE CELL ---]

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

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```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

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```
np.random.seed(42)
```

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```
df = pd.read_csv('../datasets/synthetic_linear_data.csv')
X = df[['X']].values
y = df[['y']].values
```

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```
df.head()
```

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```
df.describe()
```

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```
sns.scatterplot(x="X", y="y", data = df)
plt.title("Scatter plot of X vs Y")
plt.show()
```

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```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
```

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```
def compute_cost(X,y,w,b):
    n = len(y)
    y_pred = w * X + b
    cost = (1 / n) * np.sum((y_pred - y) ** 2)
    return cost
```

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```
def compute_gradients(X,y,w,b):
    n = len(y)
    y_pred = w * X + b
    dw = (2 / n) * np.sum((y_pred - y) * X)
    db = (2 / n) * np.sum(y_pred - y)
    return dw, db
```

[--- CODE CELL ---]

```
def gradient_descent(X, y, w, b, lr=0.01, epochs = 1000):
    cost_history = []
```

```

for _ in range(epochs):
    dw, db = compute_gradients(X, y, w, b)
    w -= lr * dw
    b -= lr * db
    cost = compute_cost(X, y, w, b)
    cost_history.append(cost)

return w, b, cost_history

```

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```

w_init = 0
b_init = 0
w,b,cost_history = gradient_descent(
    X_train.flatten(),
    y_train.flatten(),
    w_init,
    b_init,
    lr=0.01,
    epochs=1000
)
print("Learned w:", w)
print("Learned b:", b)

```

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```

y_pred_custom = w * X_test + b
y_pred_custom

```

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```

mse_custom = mean_squared_error(y_test, y_pred_custom)
r2_custom = r2_score(y_test, y_pred_custom)
print("Custom MSE:", mse_custom)
print("Custom R2:", r2_custom)

```

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```

plt.scatter(X_test, y_test, label = "Actual")
plt.plot(X_test, y_pred_custom, color="green", label="From scratch")
plt.legend()
plt.title("Linear Regression From Scratch")
plt.show()

```

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```

plt.plot(cost_history)
plt.xlabel("Epochs")
plt.ylabel("Cost (MSE)")
plt.title("Trainig Loss Curve")
plt.show()

```

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```

#cost is decresing hence we know that gradient descent is working

```

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